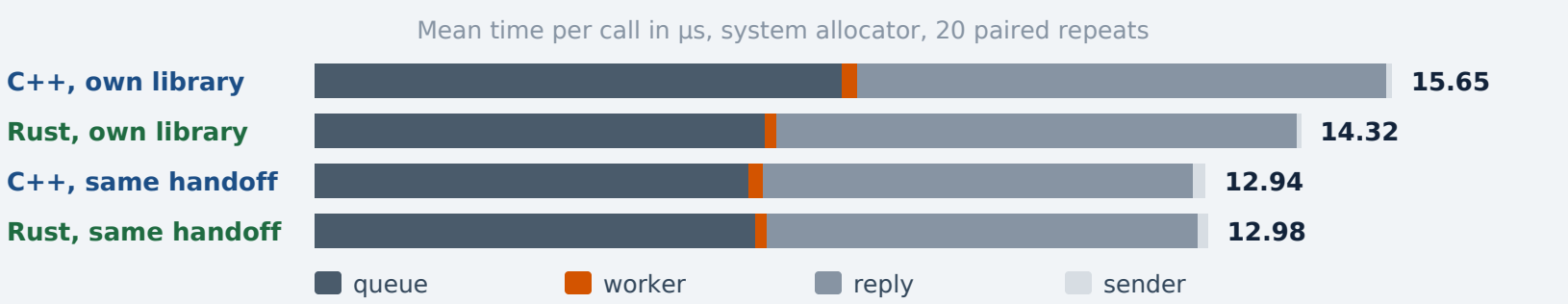
Each language's own standard library

	sender	queue	worker	reply
C++	**0.08 µs** prepares the next request	**7.67 µs** a mutex and two condition variables	**0.21 µs** runs the same ShardCore code	**7.69 µs** std::promise
Rust	**0.07 µs** prepares the next request	**6.55 µs** standard sync_channel	**0.16 µs** runs the same ShardCore code	**7.55 µs** standard channel()

One handoff written the same way in both

	sender	queue	worker	reply
C++	**0.18 µs** prepares the next request	**6.31 µs** ring behind one futex lock	**0.20 µs** runs the same ShardCore code	**6.25 µs** one slot per sender
Rust	**0.15 µs** prepares the next request	**6.41 µs** ring behind one futex lock	**0.16 µs** runs the same ShardCore code	**6.26 µs** one slot per sender

A sleeping thread is woken with one futex call. A thread that is awake is not woken.